12. With reference to the above passage, the following assumptions have been made:

1. Biotechnology regulation is an evolving process.

2. Participation of people is needed in policy decisions regarding biotechnology regulation.

3. Biotechnology regulation should take into account socio-economic aspects in decision-making.

4. Wider involvement of political executive in biotechnology regulation improves its effectiveness in dealing with the country's trade policies and international obligations.

Which of the above assumptions are valid?

(a) 1, 2 and 4 only (b) 1 and 3 only
(c) 2, 3 and 4 only (d) 1, 2 3 and 4

Ans. (b) : It can be easily inferred from the given passage that biotechnology regulation is a developing or emerging biological process, due to which the government must also take into account its socio-economic aspects before taking a decision on biotechnology regulation. Will have to keep in therefore, assumptions (1) & (3) seems to be valid.

13. Which one of the following statements best implies the crux of the passage?

(a) Precautionary principles is not given importance in current debate on developing GM crops
(b) Biotechnology is not currently used in climate change mitigation and adaptation mechanisms.
(c) Biotechnology's role is not confined to the current priorities of developing GM crops.
(d) The negative impacts of biotechnology are not properly understood.

Ans. (c) : It is clear from the passage that the role of biotechnology is not limited to the current priorities of developing GM crops in addition, domestic regulation of biotechnology should not be seen in isolation from trade and international treaties and agreements. Hence, option (c) is the correct answer.

Directions for the following 6 (six) items :

Read the following five passage and answer the items that follow. Your answers to these items should be based on the passages only.

Passage–1

Bank credit to the industrial sector has started shrinking. Its decline has been a serious concern as credit growth is essential to revive investment. The problem's origins lie in the incomplete reforms of the last 25 years. An institutional change that should have followed the 1991 reforms should have been setting up of a resolution corporation for banks. In a market economy with booms and busts, banks should be allowed to be set up and to fail. Today, we cannot shut down banks because there is no proper system to shut them down. Weak loss-making banks continue to need more capital.

14. Which one of the following is the most logical and rational inference that can be made from the above passage?

(a) Indian banking system is not able to help the country in its economic growth.
(b) Economic reforms that started in 1991 have not helped in improving the economy to expected levels.
(c) India lacks the institutional mechanism to deal with the failure of banks.
(d) Encouraging the foreign investments in our industrial sector is a good alternative to this sector's dependence on banks for credit.

Ans. (c) : The most logical and rational conclusion can be drawn from the above passage that the economic reforms started in 1991 through libertization, privatization, and globalization have not achieved the expected success in improving the condition of the economy of the nation and the universal central regulation institute and the lack of process have also made the success of the banks doubtful.

Passage–2

India has tremendous potential for solar energy. We all realize that we have to stop burning fossil fuels to meet our energy needs. But certain renewable resources are still going through their cost curves and learning curves to get the required amount of output. The Indian government has strongly committed to its targets of reducing emissions by 33 percent by 2030, and towards this is has initiated a strong push towards a gas-based economy and has also invested heavily in renewable energy. However, business houses are very of investing too heavily in renewable energy at a time when the technology is not yet ready.

15. Which one of the following is the most logical and rational inference that can be made from the above passage?

(a) India's commitment to reduce emissions by 33% is unlikely to be achieved.
(b) India should import gas rather than invest in renewable resources.
(c) Getting renewable resources to market too soon may be costly.
(d) India should put in more efforts in the exploration of natural gas.

Ans. (c) : The most rational & logical conclusion that can be drawn from the above passage is that it can be very costly to bring renewable resources to the market early due to unavailable of more cost-effective and high level of technological up-gradation at present. Hence option (c) is the correct answer.

16. With reference to the above passage, the following assumptions have been made :

1. Governments often provide inefficient and costly subsidies for technologies that may not be ready in the near future.

2. India's commitment of reducing emissions by 33% by 2030 shall be on the basis of gas-based economy.

Which of the above assumptions is/are valid?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Ans. (a) : In above passage, given statement 1 is valid because provided costly and insufficient subsidies for technologies that may not be ready in the near future.
Hence, option (a) is correct answer.

Passage-3

Genome editing is different from genome modification, Genome editing typically involves finding the part of plant genome that could be changed to render it less vulnerable to disease, yields. Researchers use 'molecular scissors' to dissect the genome and repair it, which is a process that occurs naturally when plants are under attack from diseases and can throw up new mutations that enable the plant to survive future attacks. This evolutionary process can effectively be speeded up now that it is possible to examine plant genomes in detail in laboratories, and create mechanisms through which the relevant genes can be altered very precisely.

17. With reference to the above passage, the following assumptions have been made :

1. Genome editing does not required the transfer of genes from one plant to another.

2. Through genome editing, the chosen genes can be altered precisely in a manner akin to the natural process that helps plants to adapt to the environmental factors.

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Ans. (c) : The valid assumption in the context of the passage is that genome editing does not require the transfer of genes from one plant to another. It is mentioned again in the passage that through genome editing, selected genes can be changed in a natural way, which ultimately helps the plants to adopt to environmental factors. Hence both the assumptions (1) & (2) are valid.

Passage-4

Many people understand the connection between solid waste management and health in terms of the consequences of unattended heaps of dry garbage which become home for flies and other vermin. However, there is another aspect that is not well-understood, that is, what happens when unscientific solid waste management combines with poor drainage and dumping of untreated sewage into drains which are meant to carrying storm water during rains. The result is choked drains which are full of stagnant water breeding mosquitoes, resulting in the spread of water-borne diseases.

18. In the context of India, which one of the following statements best reflects the critical message of the passage?

(a) In India, the drainage networks are not separate for sewerage and storm water.
(b) Urban local bodies do not have enough resources and legislative authority to deal with the problems of waste management.
(c) Solid waste management should be integrated with the maintenance of drainage and sewerage networks.
(d) Bad management of solid waste and sewerage systems by our municipalities is the reason for drinking water shortages in our cities.

Ans. (c) : The logical and rational message of the given passage is that solid waste management should be integrated with drainage and sewerage network management (Sewage System), as solid waste management non-functioning can result in various water borne problems and diseases, face by Urban Citizens.

Passage-5

In part III of the Constitution, which assures people certain fundamental rights, Article 25 proclaims that "all persons are equally entitled to freedom of conscience and the right freely to profess, practice and propagate religion". What people fail to notice is that this proclamation is prefixed with the words "subject to public order, morality, health and to the other provisions of this Part," which set conditions precedent for the legal protection of religious practices of any community. The closing words of this prefatory rider in Article 25 virtually constitute a subordination clause placing other fundamental rights mentioned in Part III over and above the right to religious freedom. Among those other fundamental rights is the right to equality before law and equal protection of laws-assured at the outset and elaborated in later articles to mean, inter alia, that the State shall not deny equal protection of laws to any person or group of persons on the basis of religion alone.